

BTCVC602 Foundation Engineering

Module 1: (6 Lectures)

Introduction, General requirements to be satisfied for satisfactory performance of foundations, Soil exploration: Necessity, Planning, Exploration Methods, Soil Sampling Disturbed and undisturbed, Rock Drilling and Sampling, Core Barrels, Core Boxes, Core Recovery, Field Tests for Bearing Capacity evaluation, Test Procedure & Limitations

Module 2: (7 Lectures)

Bearing Capacity Analysis - Failure Modes, Terzaghi's Analysis, Specialization of Terzaghi's Equations, Skempton Values for N_c , Meyerhof's Analysis, I.S. Code Method of Bearing Capacity Evaluation, Effect of Water Table, Eccentricity of load, Safe Bearing Capacity and Allowable Bearing Pressure, Settlement Analysis: Immediate Settlement - Consolidation Settlement, Differential Settlement, Tolerable Settlement, Angular distortion

Module 3: (5 Lectures)

Foundations for Difficult Soils - Guidelines for Weak and Compressible Soils, Expansive soil, Parameters of Expansive Soils, Collapsible Soils and Corrosive Soils, Causes of Moisture changes in Soils, Effects of Swelling on Buildings, Preventative Measures for Expansive Soils, Modification of Expansive Soils, Design of Foundation on Swelling Soils, Ground Improvement Methods: for general considerations, for Cohesive Soils, for Cohesion less Soils,

Module 4: (5 Lectures)

Shallow Foundations: Assumptions & Limitations of Rigid Design Analysis, Safe Bearing Pressure, Settlement of Footings, Design of Isolated, Combined, Strap Footing (Rigid analysis), Raft Foundation (Elastic Analysis), I. S. Code of Practice for Design of Raft Foundation

Module 5: (7 Lectures)

Deep foundations: Pile Foundation: Classification, Pile Driving, Load Carrying Capacity of Piles, Single Pile Capacity, Dynamic Formulae, Static Formulae, Pile Load Tests, Penetration Tests, Negative skin Friction, Under Reamed Piles, Group Action of Piles, Caissons Foundations: Box, Pneumatic, Open Caissons, Forces, Grip Length, Well Sinking, Practical Difficulties And Remedial Measures

Sheet Piles: Classification, Design of Cantilever Sheet Pile in Cohesionless and Cohesive soils. Design of Anchored sheet Pile by Free Earth Support Method, Cellular Cofferdams: Types, Cell Fill Stability Considerations

Module 6: (6 Lectures)

Slope Stability: Different Definitions of Factors of Safety, Types of Slope Failures, Stability of an Infinite Slope of Cohesion less Soils, Stability Analysis of an Infinite Slope of Cohesive Soils, Stability of Finite Slopes- Slip Circle Method, Semi Graphical and Graphical Methods, Friction Circle Method, Stability Number: Concept and its use

Sample question Bank for Foundation Engineering

Module – 01 [2 marks question]

Q	DESCRIPTION OF QUESTION	Marks
1	What is foundation? List four requirement of foundation	02
2	What is soil exploration? List four objectives of soil exploration	02
3	What is sample, sampler and sampling	02
4	Differentiate between representative and non-representative sample	02
5	Differentiate between disturbed and undisturbed sample	02
6	What is core recovery? If core recovery is more than 100% what you infer from it	02
7	What is rock quality designation? If RQD is 0% what do you infer from it	02
8	What is inside clearance? Why it is required for the sampler	02
9	What is outside clearance? Why it is required for the sampler	02
10	What is seating load in case of plate load test? Why it is required	02
11	What are the corrections applied to results of standard penetration test	02
12	How do you estimate the ultimate bearing capacity of plate from load settlement curve in plate load test	02
13	What is the relation between ultimate capacity of plate and that of a footing (in case of sandy soil)	02
14	What are the three conditions for applying Dilatancy corrections for SPT test	02
15	Draw neat labelled diagram of set up of SPT test	02

Module – 01 [4 marks each]

Q	DESCRIPTION OF QUESTION	Marks																			
1	Discuss the limitation of Plate load test	04																			
2	Explain the corrections to be applied on results of standard penetration test.	04																			
3	With an illustrative sketch explain how you calculate settlement of a footing in case of clayey soil and that in case of sandy soil.	04																			
4	Following observations were recorded during standard penetration test carried out on sandy soil at a depth of 9m with water table at 6m depth The saturated unit weight of soil is 20kN/m ³ and submerged unit weight is 11kN/m ³ Number of blows required for three successive 150mm penetration of the sampler were 20, 25 and 22 respectively.	04																			
5	Following observations were recorded during plate load test on sandy soil using plate of size 300mm. <table border="1"><tr><td>Load (kN)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Settlement (mm)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> Draw load settlement curve and hence calculate ultimate bearing capacity of footing of size 1.5mX1.5m	Load (kN)									Settlement (mm)									04	
Load (kN)																					
Settlement (mm)																					
6	Two sampler A and B are available at site which sampler do you use and why. Details of samplers A and B are as follows <table border="1"><tr><th rowspan="2">Sampler</th><th colspan="2">Cutting edge dimension (mm)</th><th colspan="2">Sampling tube dimension (mm)</th></tr><tr><th>External Diameter</th><th>Thickness</th><th>External diameter</th><th>Thickness</th></tr><tr><td>A</td><td>58</td><td>2.5</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td><td></td></tr></table>	Sampler	Cutting edge dimension (mm)		Sampling tube dimension (mm)		External Diameter	Thickness	External diameter	Thickness	A	58	2.5			B					04
Sampler	Cutting edge dimension (mm)		Sampling tube dimension (mm)																		
	External Diameter	Thickness	External diameter	Thickness																	
A	58	2.5																			
B																					
7	Following details were obtained during rock core sampling. Determine the rock quality designation. Length of cores obtained during sampling in sequence 86, 124, 28, 231, 65, 135, 85, 185. Also comment on quality of the sample	04																			

Module – 02 [2 marks question]

Q	DESCRIPTION OF QUESTION	Marks
1	Differentiate between shear failure and settlement failure	02
2	What do you understand by general shear failure? Write four important characteristics of general shear failure	02
3	What do you understand by local shear failure? Write four important characteristics of local shear failure	02
4	What do you understand by Punching shear failure? Write four important characteristics of punching shear failure	02
5	Write Terzaghi's Bearing capacity equation for general shear failure and explain the meaning of each term involved in the equation	02
6	Write Skempton's bearing capacity equation for general shear failure and explain the meaning of each term involved in the equation	02
7	How shear parameters of soil are modified for local shear failure	02
8	What is the effect of eccentricity of load on bearing capacity	02
9	What is the effect of water table on bearing capacity of soil	02
10	What is settlement? what are its component	02
11	What is immediate settlement? How immediate settlement is calculated	02
12	What is consolidation settlement? How it is calculated	02
13	What is differential settlement and tilt for a footing	02
14	Calculate immediate settlement for a square footing of size 2m carrying axial load of 800kN, placed in a soil having poisons ratio of 0.42 and modulus of elasticity of soil 50000kPa assume influence coefficient = 0.95	02
15	Calculate consolidation settlement of a square footing of size 2m carrying a load of 750kN placed at a depth of 2m in a soil having void ratio 0.45, compression index 0.36. If the thickness of compressible clay layer is 6m, initial pressure developed at centre of this layer is 120kN/m ² and overburden pressure at centre of clay layer is 75kN/m ²	02

Module – 02 [4 marks each]

Q	DESCRIPTION OF QUESTION	Marks
1	Write Meyerhoff's equation for bearing capacity explain meaning of each term involved in it	04
2	How shape factors are calculated in Terzaghi method and IS code method	04
3	How depth and inclination factors are calculated in IS code method	04
4	How shape and depth factors are calculated in Meyerhof method	04
5	Square footing of size 2m is placed at a depth of 2m in a soil having unit weight of 18kN/m ³ , the ultimate bearing capacity of the footing was found 750kN/m ² . What is the maximum safe load this footing will carry	04
6	Calculate safe bearing capacity for a strip footing of width 2.5m placed at a depth of 2m in a soil having $r = 19\text{kN/m}^3$, $c = 20\text{kN/m}^2$ and $\phi = 35^\circ$ Assume soil fail by general shear, water table is at large depth Use Terzaghi equation. For $\phi = 35^\circ$ Terzaghi's bearing capacity factors are $N_c = 57.8$ $N_q = 41.4$ and $N_r = 42.4$	04
7	Rectangular footing of size 3m X 2m carries axial load inclined at an angle of 10° with vertical. Footing is placed at a depth of 2m in a soil Water table was observed at a distance of 1m below footing, what are the water table correction factors and inclination factors in Meyerhof analysis	04

Module – 02 [6 marks each]

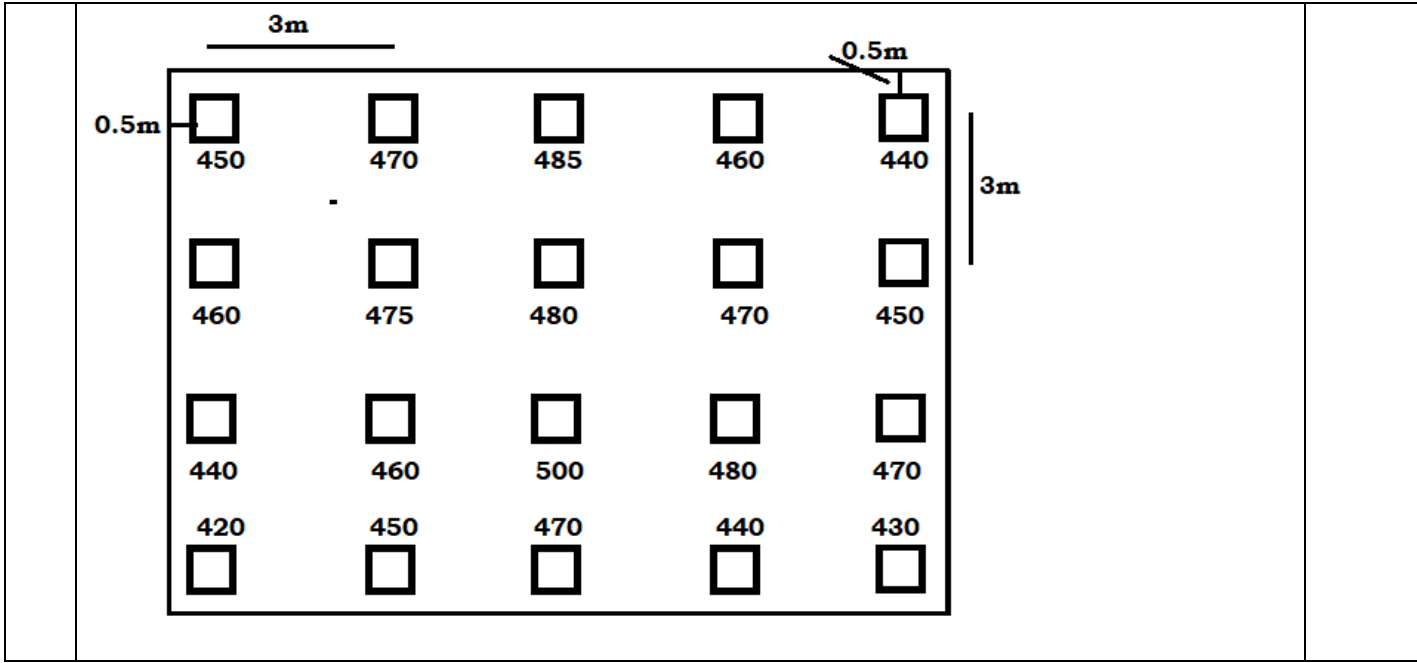
Q	DESCRIPTION OF QUESTION	Marks
1	Calculate safe bearing capacity for a square footing of width 2.0m placed at a depth of 2m in a soil having $r = 19\text{kN/m}^3$, $c = 20\text{kN/m}^2$ and $\phi = 30^\circ$ Assume soil fail by general shear, water table is at large depth Use IS code method. For $\phi = 30^\circ$ IS code bearing capacity factors are $N_c = 30.14$ $N_q = 18.4$ and $N_r = 22.4$	06
2	Design a rectangular footing with L/B ratio = 1.25 to carry safe load of 800kN. Footing is to be placed at a depth of 2m in a soil $r = 19\text{kN/m}^3$, $c = 20\text{kN/m}^2$ and $\phi = 35^\circ$ Assume soil fail by general shear, water table is at large depth Use Terzaghi equation. For $\phi = 35^\circ$ Terzaghi's bearing capacity factors are $N_c = 57.8$ $N_q = 41.4$ and $N_r = 42.4$	06
3	Calculate safe bearing capacity for a circular footing of width 2.0m placed at a depth of 2m in a soil having $r = 19\text{kN/m}^3$, $c = 36\text{kN/m}^2$ and $\phi = 0^\circ$ Assume soil fail by local shear, water table is at large depth Use Meyerhof method.	06
4	Calculate consolidation settlement for square footing of size 2m carrying a load of 800kN. Footing is placed at a depth of 2m in purely clayey soil up to a depth of 10m and then rock. Water table was at a depth of 1m below ground level. Properties of clay are ($r = 19\text{kN/m}^3$ and $r_{sat} = 21\text{kN/m}^3$, $e = 0.48$, liquid limit = 62%) Assume load dispersion 1V : 2H	06
5	Calculate safe load that can be placed on a rectangular footing at an angle of 10° with vertical using following details Size of footing – 3m X 2m Depth of footing – 1.5m Soil properties - $r = 19\text{kN/m}^3$, $c = 20\text{kN/m}^2$ and $\phi = 30^\circ$ Water table – 1m below ground level Assume soil fail by general shear Use – Meyerhof method (for $\phi = 30^\circ$, $N_c = 30.1$ $N_q = 18.4$ and $N_r = 15.7$)	06

Module – 03 [2 marks question]

Q	DESCRIPTION OF QUESTION	Marks
1	What is difficult soil? How they are grouped	02
2	What is weak and compressible soil? What are the guidelines to be followed while constructing foundation in such soil	02
3	What is collapsible soil? What are the guidelines for construction in such soil (any two)	02
4	What is corrosive soil? What are the guidelines for construction in such soil (any two)	02
5	What is expansive soil? What are the guidelines for construction in such soil (any two)	02
6	Explain any two causes of moisture change in soil	02
7	What do you understand by ground improvement? What are different techniques of ground improvement	02
8	What are the problems associated with expansive soil	02
9	Explain with neat illustrative sketches any two guidelines for construction of foundation on weak soil	04
10	Explain with neat illustrative sketches any two guidelines for construction of foundation on collapsible soil	04
11	Explain with neat illustrative sketches any two guidelines for construction of concrete foundation on corrosive soil	04
12	Explain with neat illustrative sketches any two guidelines for construction of steel foundation on corrosive soil	04
13	What are the common damage in building on expansive soil? explain any four with diagram	04
14	Explain briefly with sketch on “controlling water’s access to the soil” guidelines for construction on expansive soil	04
15	Explain briefly with sketch on “altering the soil” guideline for construction on expansive soil	04
16	Explain briefly with sketch on “Altering the method of construction” guideline for construction on expansive soil	04

Module – 04 [2 marks question]

Q	DESCRIPTION OF QUESTION	Marks									
1	With illustrative sketch explain when to provide isolated footing	02									
2	With illustrative sketch explain when to provide combined rectangular footing	02									
3	With illustrative sketch explain when to provide combined trapezoidal footing	02									
4	With illustrative sketch explain when to provide strap footing	02									
5	With illustrative sketch explain when to provide strip footing	02									
6	With illustrative sketch explain when to provide Mat footing	02									
7	With illustrative sketch explain when to provide slab type mat footing	02									
8	With illustrative sketch explain when to provide slab and beam type mat footing	02									
9	With illustrative sketch explain when to provide slab type mat footing thickened near column	02									
10	With illustrative sketch explain when to provide cellular mat footing	02									
11	What is buoyant mat? When this type of mat is used	02									
12	Explain steps involved in design of footing by Housel Method	02									
13	Following observations were recorded during plate load test conducted at a construction site at a depth of 1.8m the safe load on plate for a settlement of 25mm were as given below <table border="1"> <thead> <tr> <th>Test</th><th>Size of plate</th><th>Safe load</th></tr> </thead> <tbody> <tr> <td>1</td><td>300mm</td><td>50kN</td></tr> <tr> <td>2</td><td>600mm</td><td>94kN</td></tr> </tbody> </table> Design a square footing at this site to carry a safe load of 600kN the maximum allowable settlement for footing is 25mm	Test	Size of plate	Safe load	1	300mm	50kN	2	600mm	94kN	06
Test	Size of plate	Safe load									
1	300mm	50kN									
2	600mm	94kN									
14	Design of combined footing for two adjacent column using following details Load on each column – 800kN Size of each column – 500mm X 500mm c/c spacing of column – 3.0m SBC of soil - 100kN/m² Justify footing type you use	06									
15	Design of combined footing for two adjacent columns using following details Load on external column – 600kN Load on internal column – 800kN c/c distance between column – 3.6m Size of each column – 450mm External column is flushed with boundary SBC of soil – 200kN/m² Draw plan and elevation of footing with design dimensions	06									
16	Design of combined footing for two adjacent columns using following data Load on external column – 800kN Load on internal column – 600kN c/c distance between column – 3.6m Size of each column – 400mm Clear distance of external column from boundary – 0.2m SBC of soil – 200kN/m² Assume length of footing – 4.5m Draw plan and elevation of footing with designed dimensions	06									
17	Design a strap footing for two adjacent column using following data Load on external column – 600kN Load on internal column – 800kN c/c distance between column – 5.6m Size of each column – 400mm SBC of soil – 150kN/m² Clear distance of external column from boundary – 0.2m Draw plan and elevation of footing with designed dimensions	06									
18	With illustrative sketch explain the concept of buoyant raft	04									
19	Explain the procedure for design of Mat by Rigid method	04									
20	A Mat footing for twenty columns is as shown in figure. c/c distance between the columns is 3m in both direction and the boundary columns are at clear distance of 0.5m from the edge of mat. Determine Resultant column load on mat Position of resultant load Eccentricity of resultant load Check safety of mat against shear failure	06									



Module 05

Q	DESCRIPTION OF QUESTION	Marks
1	Write any two differences between shallow and deep foundation	02
2	Mention any four situation where you recommend pile foundation	02
3	How piles are classified discuss any four criteria for classifying pile	02
4	Differentiate between friction pile and bearing pile	02
5	Differentiate between replacement pile and displacement pile	02
6	Draw plan and elevation of pile group consisting of 12 piles arranged in four rows and three columns. Label all parts of it including pile cap	02
7	What do you understand by pile capacity? What are the different methods for finding pile capacity	02
8	What is efficiency of pile group? Discuss any two methods of finding efficiency of pile group	02
9	What is negative skin friction? How it is calculated? What are the methods of reducing negative skin friction	04
10	How the pile capacity is estimated by using static method? write meaning of all the terms used	04
11	How the pile capacity is estimated by using Dynamic method? Explain any one dynamic formula.	04
12	How the pile capacity is estimated by using Pile load test? Explain briefly with a neat labelled diagram	04
13	Explain Felds rule for finding the efficiency of pile group	04
14	Explain the procedure for construction of under-ream pile	04
15	Calculate capacity of circular pile of 400mm diameter and 10m length to be used in clayey soil having $r = 20\text{kN/m}^3$, $c = 54\text{kN/m}^2$, $\Phi = 0$ assume adhesion coefficient $\alpha = 0.60$	04
16	Calculate capacity of square pile of 500mm size and 9m length to be used in two layered clayey soil with following properties. Top layer thickness 4m, $r = 17\text{kN/m}^3$, $c = 34\text{kN/m}^2$, $\Phi = 0$ $\alpha = 0.82$, bottom layer thickness 25m, $r = 19\text{kN/m}^3$, $c = 45\text{kN/m}^2$, $\Phi = 0$ $\alpha = 0.72$.	04
17	Calculate capacity of circular pile of 600mm diameter and 8m length to be used in sandy soil having $r = 19\text{kN/m}^3$, $c = 0\text{kN/m}^2$, $\Phi = 30^\circ$ $k = 0.9$ and $\delta = 10^\circ$	04
18	Design a circular pile to carry a load of 600kN, to be used in clayey soil having $r = 19\text{kN/m}^3$, $c = 45\text{kN/m}^2$, $\Phi = 0$ assume $\alpha = 0.60$	04
19	Determine efficiency of pile group consisting of 25 piles of 500mm diameter placed at c/c spacing of 1.2m by using converse Labbare formula and Felds rule	04
20	Calculate capacity of pile group consisting of 20 piles of 400mm diameter arranged in 4 rows and 5 columns with c/c spacing of 1m. The pile group is to be used in clayey soil having $r = 20\text{kN/m}^3$, $c = 54\text{kN/m}^2$, $\Phi = 0$ assume adhesion coefficient $\alpha = 0.60$. Length of each pile is 8m	06
21	Draw labelled sketch of Anchored sheet pile wall	02
22	Determine depth of embedment for cantilever sheet pile wall using following data Height of wall – 6m Soil – Sand (above and below dredge line) Soil properties $r = 18\text{kN/m}^3$, $c = 0$ and $\Phi = 30^\circ$ Draw plan, elevation and side elevation of wall with design dimensions	04
23	Determine depth of embedment for cantilever sheet pile wall using following data Height of wall – 6m Soil above dredge line – Sand($r = 18\text{kN/m}^3$, $c = 0 \text{ kN/m}^2$ and $\Phi = 30^\circ$) Soil below dredge line – clay ($r = 18\text{kN/m}^3$, $c = 40 \text{ kN/m}^2$ and $\Phi = 0^\circ$) Draw plan, elevation and side elevation of wall with design dimensions	06
24	Determine depth of embedment for cantilever sheet pile wall using following data Height of wall – 6m Soil above dredge line – clay($r = 18\text{kN/m}^3$, $c = 30 \text{ kN/m}^2$ and $\Phi = 0^\circ$) Soil below dredge line – clay ($r = 18\text{kN/m}^3$, $c = 30 \text{ kN/m}^2$ and $\Phi = 0^\circ$) Draw plan, elevation and side elevation of wall with design dimensions	06
25	Design anchored sheet pile wall using following details Height of wall – 7.2m	06

	Anchor rod position – 1.2m below ground level Soil – Sand (above and below dredge line) Soil properties $r = 18\text{kN/m}^3$, $c = 0\text{ kN/m}^2$ and $\Phi = 30^\circ$ Draw plan, elevation and side elevation of wall with design dimensions	
26	Design anchored sheet pile wall using following details Height of wall – 7.2m Anchor rod position – 1.2m below ground level Soil above dredge line – Sand($r = 18\text{kN/m}^3$, $c = 0\text{ kN/m}^2$ and $\Phi = 30^\circ$) Soil below dredge line – clay ($r = 18\text{kN/m}^3$, $c = 40\text{ kN/m}^2$ and $\Phi = 0^\circ$) Draw plan, elevation and side elevation of wall with design dimensions	06
27	Design anchored sheet pile wall using following details Height of wall – 7.2m Anchor rod position – 1.2m below ground level Soil above dredge line – clay($r = 18\text{kN/m}^3$, $c = 40\text{ kN/m}^2$ and $\Phi = 0^\circ$) Soil below dredge line – clay ($r = 18\text{kN/m}^3$, $c = 40\text{ kN/m}^2$ and $\Phi = 0^\circ$) Draw plan, elevation and side elevation of wall with design dimensions	06
28	What is cellular cofferdam? When it is to be used? What are the ideal requirement of cell fill material	04
29	What is cofferdam? Are the ideal requirement of cofferdam	04
30	Differentiate between open caisson, box caisson and pneumatic caisson	04
31	Draw sectional elevation of open caisson and label its various parts. Explain the function of any two parts of it	04
32	Draw sectional elevation of Box caisson and label its various parts. Explain the function of any two parts of it	04
33	Draw sectional elevation of Pneumatic caisson and label its various parts. Explain the function of any two parts of it	04
34	What is tilt and shift of a well? What are limiting values of tilt and shift?	04
35	Discuss any four methods of correcting tilt of a well	04

Module no.06

Q	DESCRIPTION OF QUESTION	Marks
1	What is slope? When slopes are required	02
2	Draw labelled sketch of slope indicating various parts of slope	02
3	How slopes are classified? explain with sketches	02
4	What are different factors of safety used in slope analysis	02
5	What are different types of slope failure explain with sketches	02
6	What is rotational failure? How these failures are classified	02
7	What are different methods of analysis of finite slope	02
8	Analyse infinite slope shown below. Using following details Soil – clay ($r = 18\text{kN/m}^3$, $c = 40\text{ kN/m}^2$ and $\Phi = 0^\circ$) Slope angle – 25° Height of slope – 7m	02
9	Analyse infinite slope shown below. Using following details Soil – sand ($r = 18\text{kN/m}^3$, $c = 0\text{ kN/m}^2$ and $\Phi = 35^\circ$) Slope angle – 25° Height of slope – 7m	04
10	Analyse infinite slope shown below. Using following details Soil – C – Φ Slope angle – 25° ($r = 18\text{kN/m}^3$, $c = 20\text{ kN/m}^2$ and $\Phi = 30^\circ$) Height of slope – 7m	04
11	Analyse infinite slope shown below. Using following details Soil – silt Slope angle – 25° ($r = 18\text{kN/m}^3$, $r_{sat} = 21\text{ kN/m}^3$, $c = 20\text{ kN/m}^2$, $\Phi = 30^\circ$) Height of slope – 7m Water table – 2m below ground level Seepage is parallel to slope	06
12	Analyse infinite slope shown below. Using following details Soil – sand Slope angle – 25° ($r = 18\text{kN/m}^3$, $r_{sat} = 21\text{ kN/m}^3$, $c = 0\text{ kN/m}^2$ and $\Phi = 30^\circ$) Height of slope – 7m Water table – 2m below ground level Seepage is parallel to slope	04
13	Analyse infinite slope shown below. Using following details Soil – clay Slope angle – 25° ($r = 18\text{kN/m}^3$, $r_{sat} = 21\text{ kN/m}^3$, $c = 40\text{ kN/m}^2$ and $\Phi = 0^\circ$) Height of slope – 7m Water table – 2m below ground level Seepage is parallel to slope	04
14	Determine factor of safety of slope of purely cohesive soil having cohesion is 30kN/m^2 , $r = 17\text{kN/m}^3$ $\phi = 0$, radius of slip surface = 10.8m, angle subtended at center = 115° . The C.G. of wedges is horizontally 3.5m away from Centre of slip surface. Take area of wedge 124m^2 .	04
15	An infinite slope of 5m thickness has following properties slope angle (with horizontal) = 28° . $\phi = 25^\circ$, $C = 15\text{kN/m}^2$, and unit weight of soil = 18kN/m^3 . Investigate the stability of slope	04
16	The unit weight of a soil of 30° slope is 17.5kN/m^3 , The shear parameters 'C' and ' ϕ ' for the soil are 10kN/m^2 at 20° respectively. Given that height of slope is 12m and the stability number obtained from the chart is 0.025, compute the factor of safety.	04
17	Determine factor of safety of slope of purely cohesive soil having cohesion is 30kN/m^2 , $\gamma = 17\text{kN/m}^3$ $\phi = 10^\circ$, radius of slip surface = 10.8m, angle subtended at center = 115° . The C.G. of wedges is horizontally 3.5m away from Centre of slip surface. Take area of wedge 124m^2 . Use friction circle method	

- 18** Determine the FOS of slope inclined at 30° by method of slices. Take circular failure surface passing through toe of slope with Centre at intersection of directional angles $\alpha = 26^\circ$ and $\beta = 35^\circ$. Take unit weight of soil as 19 KN/m^3 . $\phi = 25^\circ$, $C = 15 \text{ KN/m}^2$. Height of slope is 8 m . Take slices width $= 2.5 \text{ m}$.

